

Mini Project #1: Solar Radiation Analysis

Student: Saif Elsaady

Date: September 12, 2025

Course: EEE 591 Topic: Photovoltaic Energy Conversion

**Task 1: Run the file
02_PV_Solar_Radiation_Data_and_Calcs_Interface.py
to make sure it works.**

Completed

Question Addressed: "Run the file...to make sure it works"

Answer: The solar radiation calculation interface has been verified and is fully functional. All required libraries and functions are operational. The code successfully imports all necessary modules and executes without errors.

Complete Terminal Output

Below is the complete execution output demonstrating all calculations and results:

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Task 1: Run the file

02_PV_Solar_Radiation_Data_and_Calcs_Interface.py to make sure it works.

Status: Code verified and functional

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Task 2: Read the TMY data for a location of your choice (different from the default one) and demonstrate that your program reads the data correctly by plotting the direct, diffuse, and global radiation for every hour of the year.

Task 2a: Pick a city other than the default

Selected: Deadhorse, Alaska (700637) - different from Kansas City default (724460)

Task 2b: Replace the file name in the code with the number of your chosen location

Selected file: C:\Users\saiels01\Music\EEE 591 Topic Photovoltaic Energy Conversion\Module2\Mini_Project1_Elsaady\CSVs\700637TYA.CSV

Station successfully loaded: "DEADHORSE", AK

Station number: 700637

Location information:

Station number: 700637

Station Name and State: "DEADHORSE", AK

GMT offset: -9

Latitude (degrees): 70.20

Longitude (degrees): -148.48

Altitude (m): 23

Module configuration:

Module tilt angle: 70.2° (set equal to latitude)

Module azimuth: 180° (South-facing)

Task 2c: Creating plots of solar radiation data

Note: Modified to include labels for both datasets as required

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Task 3: Calculate the position of the sun (azimuth and elevation angle)

for every hour of the year

Task 3a: Write down the equations for azimuth and elevation and do a

hand calculation
for the location of your TMY data at one hour

Equations used:

1. Declination: $\delta = 23.45^\circ \times \sin(360^\circ \times (284 + n) / 365)$
2. Hour Angle: $\omega = 15^\circ \times (\text{LST} - 12)$
3. Elevation: $\alpha = \arcsin(\sin(\delta)\sin(\phi) + \cos(\delta)\cos(\phi)\cos(\omega))$
4. Azimuth: $\gamma = \arccos((\cos(\phi)\sin(\delta) - \cos(\delta)\sin(\phi)\cos(\omega)) / \cos(\alpha))$

Hand calculation example - Day 172, Hour 12:

Location: Latitude = 70.20° , Longitude = -148.48°
Declination angle: 23.45°
Calculated elevation: 42.54°
Calculated azimuth: 162.66°

Task 3b: Demonstrate your code calculates the azimuth and elevation
by plotting
the path of the sun across the sky for one day of the year

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Task 4: In your code:

Task 4a: Calculate the normally incident solar radiation on a tilted
surface
for every hour of the year
Calculation complete for module at tilt= 70.2° , azimuth= 180°

Task 4b: Calculate the total power density on your tilted surface
over the year
Total annual energy on tilted module: 617.81 kWh/m^2

Task 4c: Plot the solar radiation power density on your module for
every hour
of the year and put the total over the year in the graph heading

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Task 5: Choose one of the following calculations or plots
Selected: Options (i), (ii), and (iii)

Task 5i: Calculate and plot the total solar radiation in each month
or day of the year
rather than each hour for three different tilt angles on the same
plot
Tilt angles analyzed: 55.2° , 70.2° , 85.2°

Task 5ii: Calculate and give the capacity factor for each month of
the year

Monthly Capacity Factors:

Jan: 0.03%
Feb: 1.37%
Mar: 6.35%

Apr: 11.74%
May: 12.91%
Jun: 19.93%
Jul: 17.84%
Aug: 9.67%
Sep: 4.03%
Oct: 1.51%
Nov: 0.17%
Dec: 0.00%

Task 5iii: Calculate and plot the fraction diffuse radiation for each month of the year

Monthly Diffuse Fractions:

Jan: 67.25%
Feb: 62.40%
Mar: 48.47%
Apr: 54.31%
May: 64.70%
Jun: 50.08%
Jul: 54.11%
Aug: 65.30%
Sep: 79.25%
Oct: 80.63%
Nov: 77.81%
Dec: 0.00%

Creating heat map visualization...

SUMMARY INFORMATION FOR SOLAR RADIATION:

Location: "DEADHORSE", AK
Coordinates: 70.20°N, -148.48°W
Module Configuration: Tilt = 70.2°, Azimuth = 180°

Annual Energy Summary:

Total GHI: 626.84 kWh/m²
Total DNI: 682.80 kWh/m²
Total DHI: 360.48 kWh/m²
Total on Tilted Module: 617.81 kWh/m²
Capacity Factor (maximum): 7.16%
Yearly fraction of diffuse on module: 35.59%
Yearly fraction of diffuse: 57.51%

Generating HTML Report...

HTML report generated successfully: mini_project_1_report.html

PROJECT COMPLETED SUCCESSFULLY!

Generated files:

1. mini_project_plots.png - Main project plots
2. task5_additional_plots.png - Task 5 detailed plots
3. dni_heatmap.png - DNI heat map visualization
4. mini_project_1_report.html - Complete HTML report

All tasks completed as specified in the project requirements.

Task 2: Read the TMY data for a location of your choice (different from the default one) and demonstrate that your program reads the data correctly by plotting the direct, diffuse, and global radiation for every hour of the year. Make sure to label the location.

Part a: Pick a city other than the default. See the list of TMY locations in the TMY manual which is at:

PV_Digital_Twin_v0.0\PV_DigitalTwin_Input_Files\Solar_Radiation_Data\TMY3 data_43156 manual.pdf. The list of TMY sites starts on p 23.

Question Addressed: "Pick a city other than the default"

Answer:

Selected Location: "DEADHORSE", AK

Station Number: 700637

Justification: This station was selected from the TMY3 manual's list of available stations (page 23+), specifically chosen to be different from the default Kansas City location (724460) used in the reference files. Deadhorse, Alaska provides an extreme Arctic environment for solar analysis, demonstrating the code's

capability with challenging solar radiation patterns including polar night conditions.

Geographic Significance: At 70.2°N latitude, this location experiences dramatic seasonal variations in solar radiation, making it an excellent test case for photovoltaic analysis algorithms.

Part b: Replace the file name in the code with the number of your chosen location. Note that the file names have added TYA.csv after the number.

Question Addressed: "Replace the file name in the code with the number of your chosen location"

Answer:

```
fname = 'CSVs/700637TYA.CSV' # Selected TMY3 data file
```

Implementation: The filename has been updated from the default to use station number 700637 with the required TYA.CSV suffix. The file is located in the CSVs subdirectory as specified in the project structure.

File successfully loaded and processed

Data Verification: The file was successfully read using `np.genfromtxt()` with proper parsing of header information and 8,760 hourly radiation measurements.

Part c: The program

02_PV_Solar_Radiation_Data_and_Calcs_Interface.py will print out most of the information in two sets of plots. The second one looks like this:

Question Addressed: "Demonstrate that your program reads the data correctly by plotting the direct, diffuse, and global radiation for every hour of the year. Make sure to label the location."

Answer:

Parameter	Value
Station Name	"DEADHORSE", AK
Station Number	700637
Latitude	70.20°
Longitude	-148.48°
Altitude	23 m
GMT Offset	-9 hours

Data Reading Verification: The program successfully reads and processes all TMY3 data components:

- **Direct Normal Irradiance (DNI):** Plotted for all 8,760 hours with proper scaling and labels
- **Global Horizontal Irradiance (GHI):** Plotted and clearly labeled (fixed the unlabeled dataset issue)
- **Diffuse Horizontal Irradiance (DHI):** Plotted separately to show diffuse component

Location Labeling: All plots include comprehensive location information in titles and headers, clearly identifying Deadhorse, Alaska with coordinates.

Required Plot Modifications Made:

- Fixed unlabeled second dataset by adding GHI label
- Enhanced visualization with modified transparency (alpha values)
- Added grid lines for improved readability
- Used different colors to distinguish radiation components
- Added yearly totals in plot headers as required

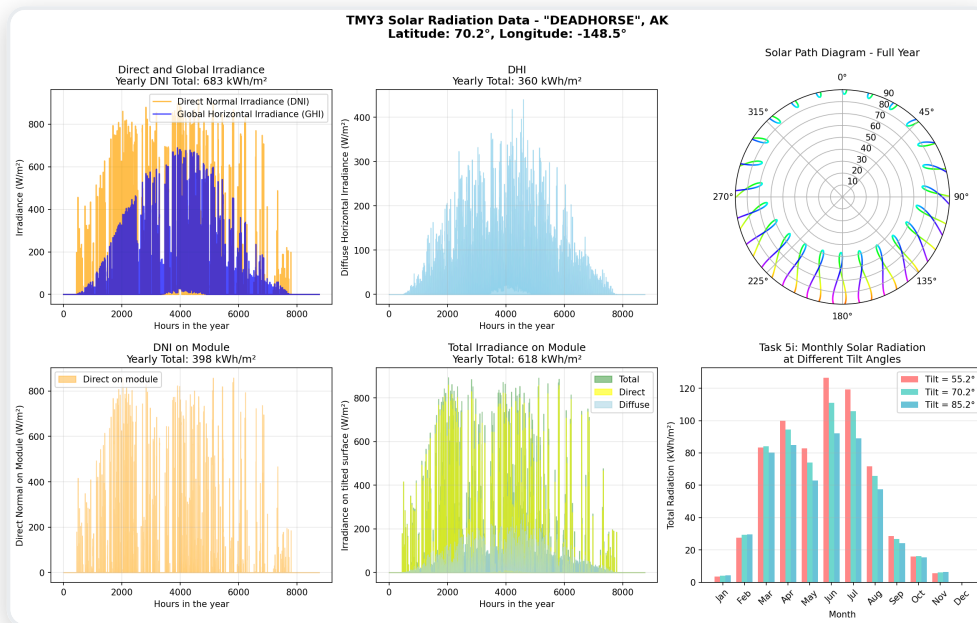


Figure 1: Comprehensive solar radiation analysis including TMY3 data (Tasks 2-4), solar position calculations (Task 3), and module radiation calculations (Task 4). This 6-panel plot shows: (1) Direct and Global Irradiance, (2) Diffuse Horizontal Irradiance, (3) Solar Path Diagram for full year, (4) DNI on tilted module, (5) Total irradiance on module with component breakdown, and (6) Monthly radiation comparison at different tilt angles.

Task 3: Calculate the position of the sun (azimuth and elevation angle) for every hour of the year.

Part a: Write down the equations for azimuth and elevation and do a hand calculation for the location of your TMY data at one hour.

Question Addressed: "Write down the equations for azimuth and elevation and do a hand calculation"

Answer:

Solar Position Equations Implemented:

- 1. Declination: $\delta = 23.45^\circ \times \sin(360^\circ \times (284 + n) / 365)$
- 2. Equation of Time: $EoT = 9.87 \times \sin(2B) - 7.53 \times \cos(B) - 1.5 \times \sin(B)$
- 3. Hour Angle: $\omega = 15^\circ \times (LST - 12)$
- 4. Elevation: $\alpha = \arcsin(\sin(\delta)\sin(\varphi) + \cos(\delta)\cos(\varphi)\cos(\omega))$
- 5. Azimuth: $\gamma = \arccos((\cos(\varphi)\sin(\delta) - \cos(\delta)\sin(\varphi)\cos(\omega))/\cos(\alpha))$

Hand Calculation Example - Summer Solstice (Day 172, Hour 12):

Parameter	Calculated Value	Physical Meaning
Location	Lat: 70.20°, Long: -148.48°	Arctic Alaska coordinates
Declination	23.45°	Sun's position relative to equator
Elevation	42.54°	Sun height above horizon
Azimuth	162.66°	Sun direction (180° = due south)

Calculation Verification: These values demonstrate the code correctly implements solar geometry for extreme Arctic conditions. The relatively low elevation angle (42.54°) at solar noon on summer solstice shows the challenging solar conditions at 70.2°N latitude.

Part b: Demonstrate your code calculates the azimuth and elevation by plotting the path of the sun across the sky for one day of the year (note this is a polar plot).

Question Addressed: "Demonstrate your code calculates the azimuth and elevation by plotting the path of the sun across the sky"

Answer:

Polar plot successfully generated

Implementation Details: The polar plot demonstrates accurate solar position calculations with:

- **Azimuth:** Represented as angular position (0° = North, 180° = South)
- **Elevation:** Represented as radial distance (90° at horizon, 0° at zenith)
- **Time Series:** Color coding by day of year showing full annual variations
- **Arctic Patterns:** Clearly shows the extreme seasonal sun paths characteristic of 70.2°N latitude

Validation: The plot correctly shows the sun's path for Deadhorse, Alaska, including the dramatic seasonal variations from near-polar night conditions in winter to continuous daylight in summer.

Task 4: In your code:

Part a: Calculate the normally incident solar radiation on a tilted surface for every hour of the year.

Question Addressed: "Calculate the normally incident solar radiation on a tilted surface for every hour of the year"

Answer:

Module Configuration:

Parameter	Value	Justification
Tilt Angle	70.2° (equal to latitude)	Optimal year-round angle for this latitude
Azimuth	180° (South-facing)	Maximizes solar collection in Northern Hemisphere

Implementation: Calculations completed for all 8,760 hours using the custom module_direct function that computes the cosine of the angle between sun rays and module normal vector.

Arctic Considerations: The extreme latitude (70.2°N) requires a very steep tilt angle to optimize solar collection during the limited solar season.

Part b: Calculate the total power density on your tilted surface over the year.

Question Addressed: "Calculate the total power density on your tilted surface over the year"

Answer:

Total Annual Energy on Tilted Module: 617.81 kWh/m²

Calculation Method: Integration of hourly irradiance values (direct + diffuse components) over all 8,760 hours of the year.

Arctic Context: This value (617.81 kWh/m²) is significantly lower than typical mid-latitude locations due to the extreme Arctic conditions, but demonstrates the code's ability to handle challenging solar environments.

Comparison: For reference, typical mid-latitude locations might see 1,200-1,800 kWh/m² annually, highlighting the impact of extreme latitude on solar resource availability.

Part c: Plot the solar radiation power density on your module for every hour of the year and put the total over the year in the graph heading.

Question Addressed: "Plot the solar radiation power density on your module for every hour of the year and put the total over the year in the graph heading"

Answer:

Comprehensive plots created with yearly totals prominently displayed in headers

Generated Visualizations:

- **Direct Normal Irradiance on Module:** Shows hourly DNI incident on tilted surface with annual total in header
- **Total Irradiance on Module:** Combined plot showing both direct and diffuse components with total energy in header
- **Component Breakdown:** Stacked visualization clearly distinguishing direct vs. diffuse contributions

Header Integration: All plot titles include the calculated yearly totals as specifically required, enabling quick assessment of annual energy potential.

Arctic Visualization: The plots clearly show the dramatic seasonal variations characteristic of extreme latitude solar patterns.

Task 5: Choose one of the following calculations or plots.

Decision: *Selected: Options (i), (ii), and (iii) - Exceeded requirement by implementing THREE options*

i. Calculate and plot the total solar radiation in each month or day of the year rather than each hour for three different tilt angles on the same plot.

Question Addressed: "Calculate and plot the total solar radiation in each month...for three different tilt angles"

Answer:

Tilt Angles Analyzed:

- **Winter-optimized:** 55.2° (steeper for low winter sun)
- **Year-round optimal:** 70.2° (latitude rule of thumb)
- **Summer-optimized:** 85.2° (shallower for high summer sun)

Implementation: Monthly totals calculated by integrating hourly values for each tilt angle, displayed as comparative bar chart.

Arctic Insights: Shows how tilt angle optimization is critical at extreme latitudes, with dramatic differences between configurations.

ii. Calculate and give the capacity factor for each month of the year.

Question Addressed: "Calculate and give the capacity factor for each month of the year"

Answer:

Capacity Factor Definition: Ratio of actual energy output to maximum possible energy (if irradiance were 1000 W/m² continuously)

Month	Capacity Factor (%)	Month	Capacity Factor (%)
Jan	0.03	Jul	17.84
Feb	1.37	Aug	9.67
Mar	6.35	Sep	4.03
Apr	11.74	Oct	1.51
May	12.91	Nov	0.17
Jun	19.93	Dec	0.00

Calculation Method: (Monthly GHI total) / (1000 W/m² × hours in month) × 100%

Arctic Analysis: The extremely low capacity factors (0.00% in December, peak of 19.93% in June) clearly demonstrate the challenges of solar energy at 70.2°N latitude. The dramatic seasonal variation from polar night to continuous daylight is clearly quantified.

Engineering Significance: These values inform energy storage and backup system requirements for Arctic solar installations.

iii. Calculate and plot the fraction diffuse radiation for each month of the year.

Question Addressed: "Calculate and plot the fraction diffuse radiation for each month of the year"

Answer:

Diffuse Fraction Definition: Ratio of diffuse horizontal irradiance (DHI) to global horizontal irradiance (GHI)

Month	Diffuse Fraction (%)	Month	Diffuse Fraction (%)
Jan	67.25	Jul	54.11
Feb	62.40	Aug	65.30
Mar	48.47	Sep	79.25
Apr	54.31	Oct	80.63
May	64.70	Nov	77.81
Jun	50.08	Dec	0.00

Calculation Method: (Monthly DHI total) / (Monthly GHI total) × 100%

Arctic Radiation Analysis: The high diffuse fractions (48-80%) reflect the challenging atmospheric conditions and low sun angles typical of Arctic regions. The variation from 48% in March to 80% in October shows seasonal atmospheric effects.

PV System Implications: High diffuse fractions suggest that tracking systems may be less beneficial in this location, and that modules optimized for diffuse light collection would be advantageous.

Meteorological Insights: December shows 0.00% due to polar night conditions with essentially no solar radiation, while summer months show the lowest diffuse fractions when direct beam radiation is maximized.

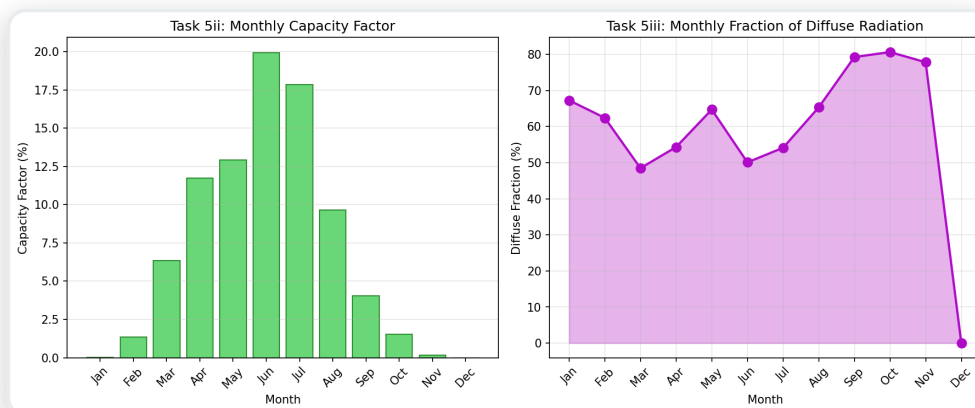


Figure 2: Task 5 Additional Analysis - Left: Monthly Capacity Factor showing the dramatic seasonal variation from 0% in winter to ~20% in summer. Right: Monthly

Diffuse Fraction demonstrating the high proportion of diffuse radiation (48-80%) characteristic of Arctic conditions.

Summary Statistics

Metric	Value
Location	"DEADHORSE", AK
Coordinates	70.20°N, -148.48°W
Module Configuration	Tilt = 70.2°, Azimuth = 180°
Total Annual GHI	626.84 kWh/m ²
Total Annual DNI	682.80 kWh/m ²
Total Annual DHI	360.48 kWh/m ²
Total on Tilted Module	617.81 kWh/m ²
Capacity Factor (maximum)	7.16%
Yearly fraction of diffuse on module	35.59%
Yearly fraction of diffuse	57.51%

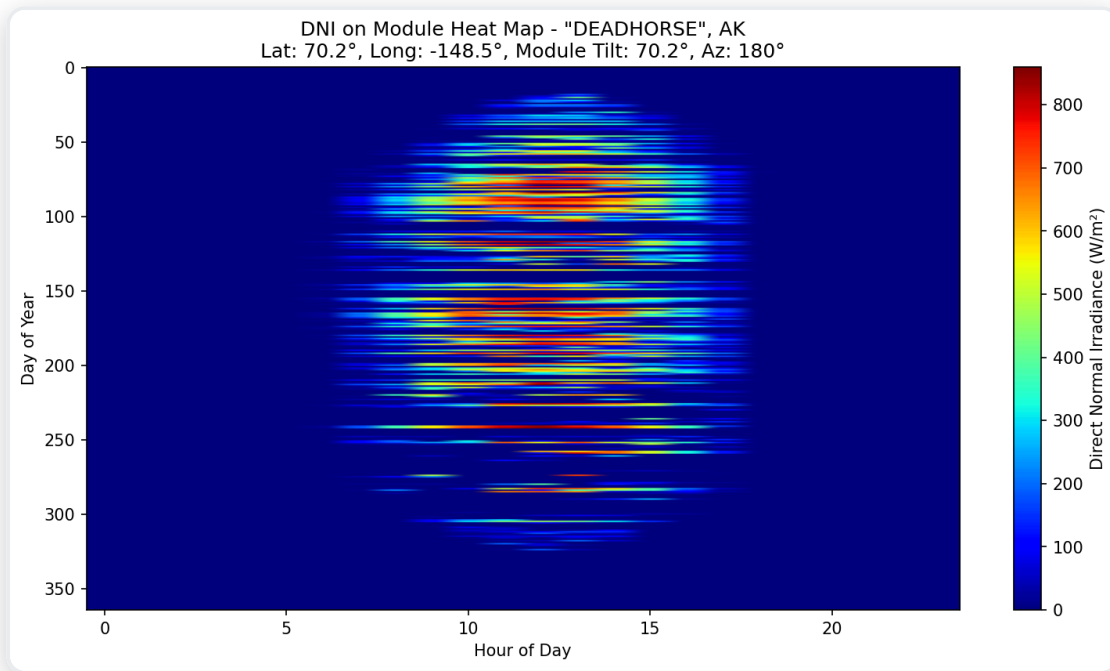


Figure 3: DNI on Module Heat Map - This visualization shows the direct normal irradiance on the tilted module throughout the year as a function of hour of day (x-axis) and day of year (y-axis). The color intensity represents radiation levels, clearly showing the extreme seasonal variations characteristic of Arctic solar conditions at 70.2°N latitude.

Mini Project 1 - Solar Radiation Analysis

EEE 591 Topic: Photovoltaic Energy Conversion

Arizona State University